

Abstract

The proposed project is an **AI-assisted investigation and evidence analysis platform** designed to address the difficulties investigators face while handling large amounts of evidence collected from different sources. During an investigation, information may be available in the form of CCTV records, GPS data, phone calls, messages, emails, financial transactions, vehicle records, access logs, witness information, and forensic documents. Since these sources can have different formats and structures, investigators often need to examine and compare them manually. This makes it difficult and time-consuming to identify connections between people, locations, events, communications, and other pieces of evidence.

To solve this problem, the proposed system brings different types of investigative evidence into a **single structured platform**. The system extracts information from different file formats and normalizes it into a common structure before storing it in a Oracle **SQL relational database**. The database organizes important investigative information such as people, locations, events, communications, vehicles, transactions, relationships, and evidence. This allows information from different sources to be connected and analyzed together rather than being examined separately.

The system also provides **predefined SQL investigation queries** for answering common investigative questions. These queries can be used to find communication between individuals, identify common locations, examine financial activity, analyze events within specific time periods, and discover relationships between different entities. This provides investigators with a structured and explainable method of analyzing the stored evidence.

An automated **correlation engine** is used to identify possible connections between entities and different pieces of evidence. In addition, an **AI-assisted component** helps interpret investigator questions and route them to the appropriate analytical modules. The results are presented through an interactive interface containing relationship graphs, timelines, location information, evidence views, and pop-up windows for displaying query results.

The system follows the principle “**AI assists, humans decide.**” It is not intended to replace investigators or make final investigative decisions. Instead, it assists them by organizing large amounts of evidence, reducing repetitive manual analysis, identifying possible relationships, and presenting relevant information in an understandable manner. The final interpretation and decision remain with the human investigator.

Overall, the proposed platform combines **evidence ingestion, data normalization, relational database management, SQL-based investigation, automated correlation, AI-assisted analysis, and an interactive user interface** into a single system. The aim is to make investigative data easier to manage and analyze, reduce the effort required to examine information from multiple sources, and help investigators discover meaningful connections and patterns in a more systematic and efficient manner.